ASEAN
GEOAI FUSION 2026

Module 4 Session 1: Introduction to Telecommunications Applications in GeoAI

AGAIF 2026 Boot Camp 1
Foundation of GeoAI

Thursday 9 July 2026

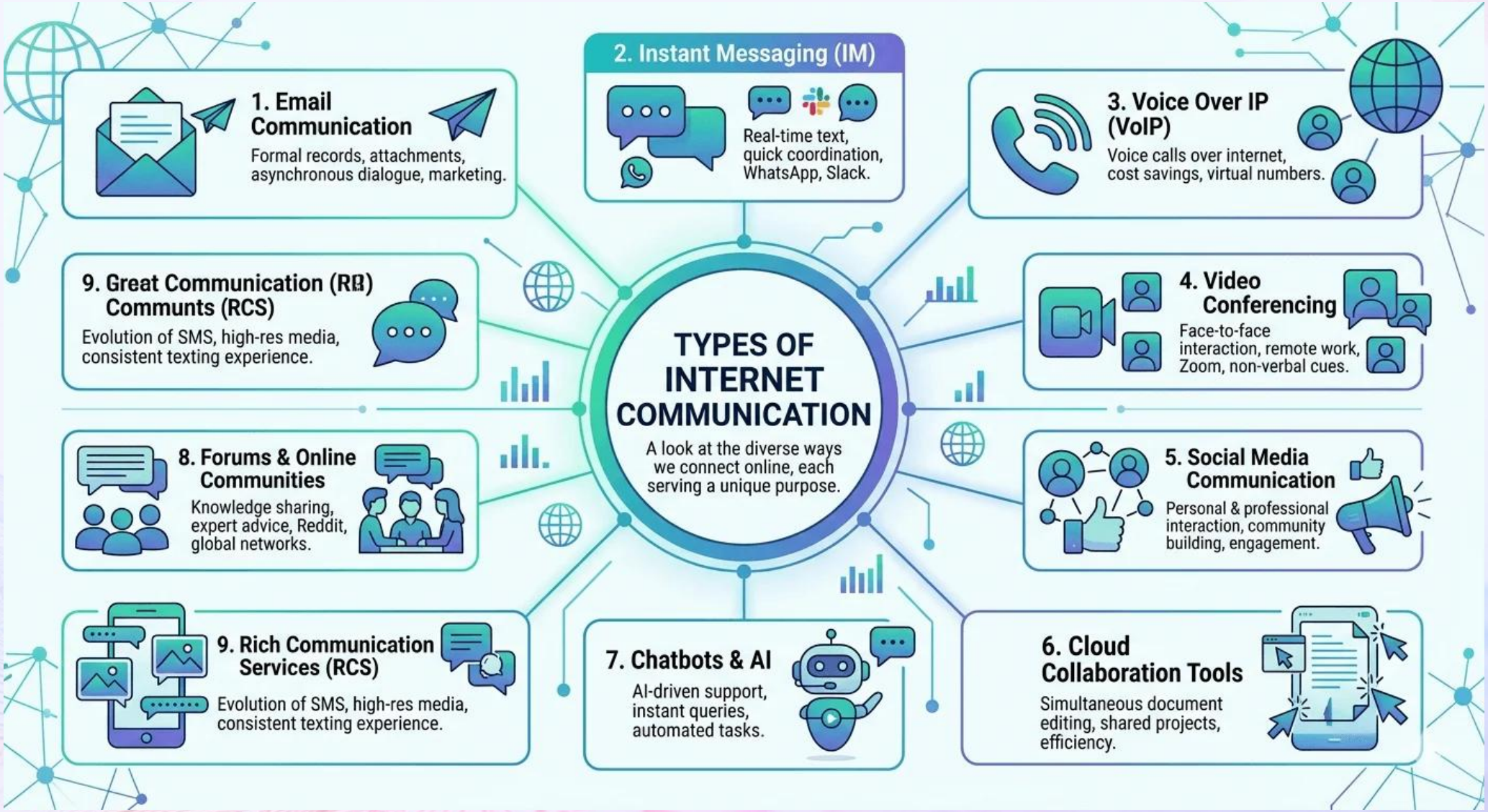
Delivered by:
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Universiti Tenaga Nasional (UNITEN)

Learning Outcomes

By the end of this session, participants should be able to:

1. Explain the role of telecommunications in modern digital systems.
2. Describe how telecommunications support GeoAI.
3. Identify applications of telecommunications in different sectors.

Why Telecommunications?



Telecommunications as the Backbone of GeoAI

- GeoAI requires two things:
  ✓ Geospatial Data ✓ Connectivity

- Without telecommunications:

1. Sensors cannot transmit data
2. Drones cannot send images
3. Satellites cannot deliver observations
4. AI cannot receive real-time information

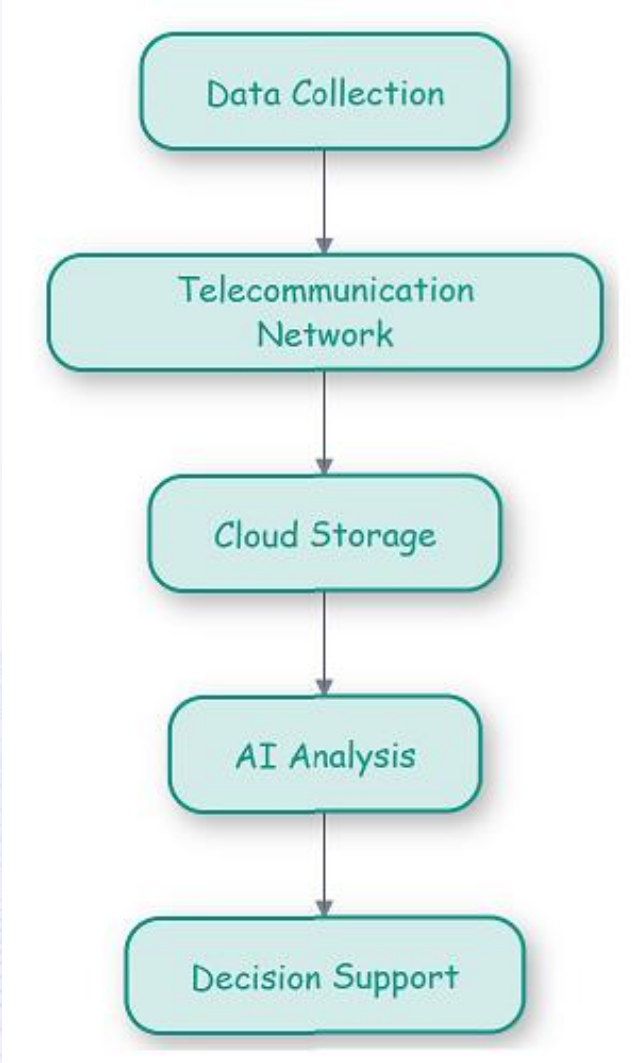
Telecommunications make GeoAI possible.

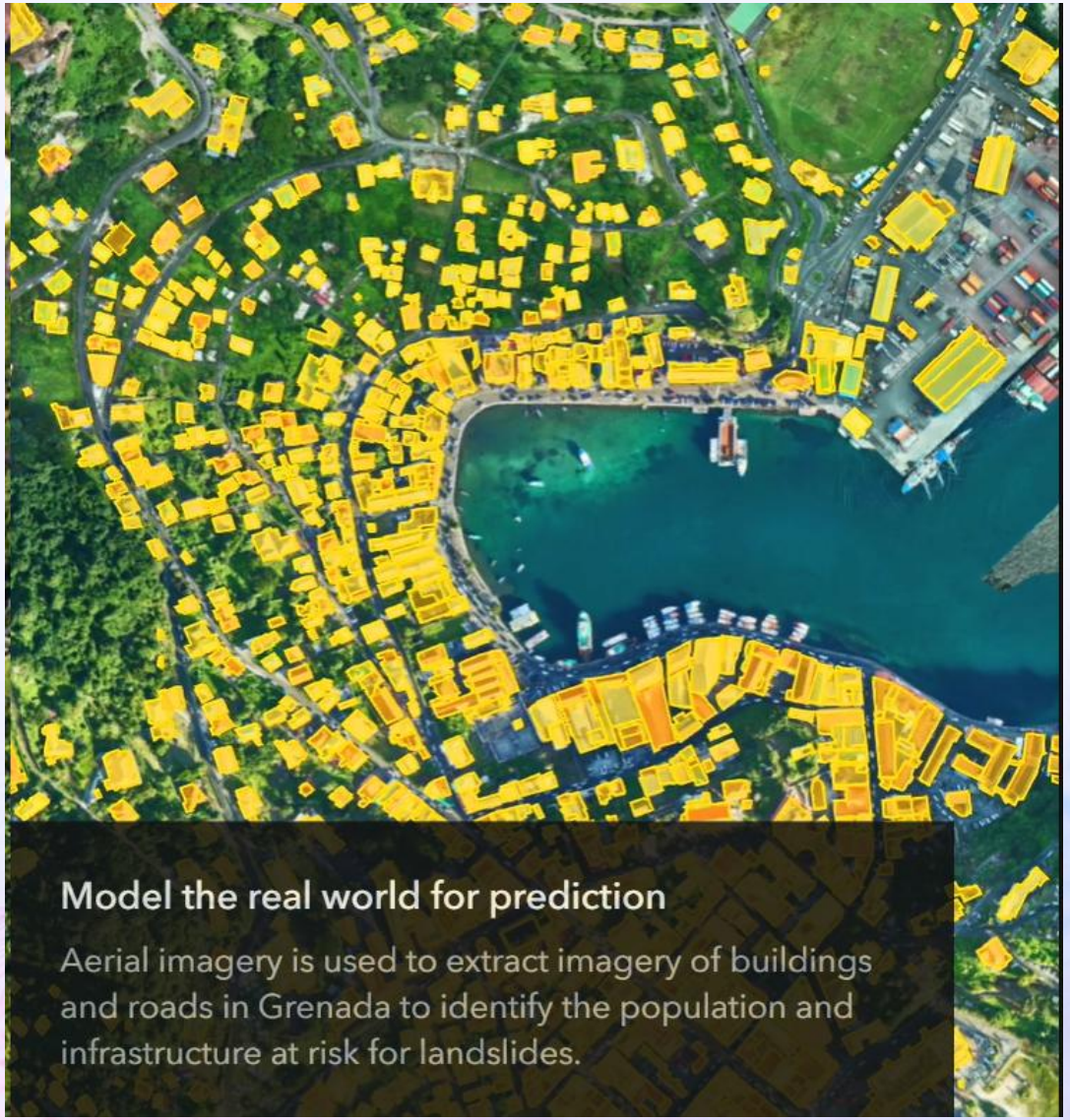
Where Does GeoAI Data Come From?



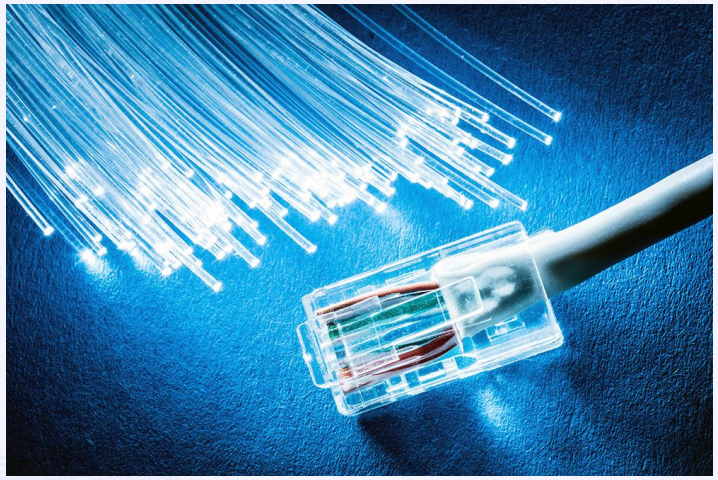
All these devices continuously collect spatial data

How Data Reaches GeoAI





Telecommunications Technologies





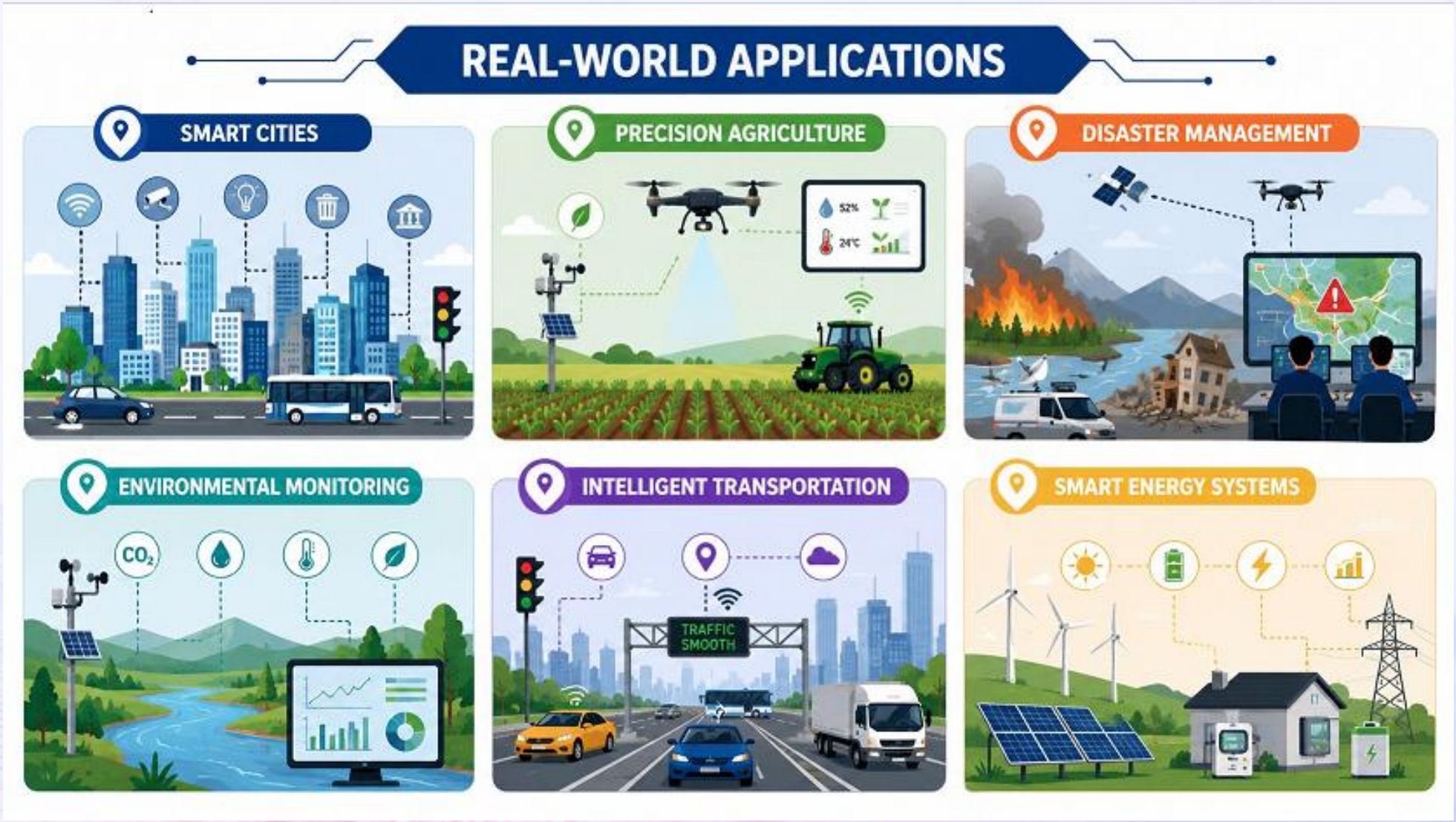








Real-World Applications



ASEAN USE CASES

Discuss examples relevant to the region:

- Flood monitoring
- Forest fire detection
- Coastal monitoring
- Rice field management
- Urban traffic monitoring
- Biodiversity conservation

Explain how telecommunications support each application.

Future Trends

EMERGING TECHNOLOGIES

Powering a Smarter, Faster, and More Connected Future

1. 5G & BEYOND



- Ultra-high speed and low latency
- Massive device connectivity
- Enabling immersive experiences
- Foundation for 6G and future innovations

2. INTERNET OF THINGS (IoT)



- Connects billions of devices
- Real-time data collection and monitoring
- Drives automation and smarter decision-making

3. ARTIFICIAL INTELLIGENCE



- Enables machines to learn and reason
- Powers automation and personalization
- Improves efficiency and accuracy
- Drives innovation across industries



CONNECTED INTELLIGENT FUTURE

4. EDGE COMPUTING



- Processes data closer to the source
- Reduces latency and bandwidth use
- Enhances real-time decision-making
- Supports IoT, AR/VR, and mission-critical applications

5. DIGITAL TWINS



- Virtual replicas of physical assets or systems
- Enables simulation, monitoring and prediction
- Improves performance and reduces operational risks

6. 6G NETWORKS



- Next-generation connectivity
- Ultra-massive capacity & ultra-low latency
- AI-native, secure, and sustainable
- Enabling human-centric intelligent world

HIGH SPEED | MASSIVE CONNECTIONS | INTELLIGENT AUTOMATION | SECURE & RELIABLE | SUSTAINABLE FUTURE

These technologies, together, are building the foundation for a smarter, more connected planet.

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Key Takeaways

✓ Telecommunications connect people, devices and sensors.

✓ Telecommunications enable real-time geospatial data sharing.

✓ GeoAI depends on reliable communication networks.

✓ Telecommunications support smarter and faster decision-making.

Without telecommunications, there is no real-time GeoAI.

Question
"How can telecommunications improve GeoAI applications in your country?"

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THANK YOU

Bootcamp and Hackathon for ASEAN Geospatial Innovation 2026